Pituitary Abnormalities

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Block: Endocrine/Urology/Reproductive

Thread: Clinical Medicine

Conflict of interest: Dr. Simon has no conflicts of interest to disclose

Further Reading

Higham, C.E., Johannsson, G., Shalet, S.M. Hypopituitarism. *The Lancet*. 2016;388(10058):2403-2415. doi:10.1016/S0140-6736(16)30053-8 (View Online Here)

Molitch, Mark E. Diagnosis and Treatment of Pituitary Adenomas: A Review. *JAMA*. 2017;317(5):516-524. doi:10.1001/jama.2016.19699 (View Online Here)

Shapiro M, Weiss JP. Diabetes Insipidus: A Review. *Journal of Diabetes and Metabolism*. 2012; S6:009. (View Online Here)

Learning Objectives

By exam time, you should be able to:

1. Describe basic pituitary anatomy and physiology
2. Identify and explain the clinical syndromes caused by pituitary adenomas and pituitary hormone hypersecretion and the basic approach to diagnosis
3. Define and recognize the clinical syndrome of hypopituitarism
4. Summarize the clinical syndromes of diabetes insipidus (AVP deficiency) and SIADH
5. Review the basic approach to the pituitary incidentaloma

Study Questions

1. A 38 year-old female with a history of poorly controlled diabetes and hypertension is referred to an endocrinologist for management of her blood sugar. During her physical exam, she is noted to have a round face, supraclavicular fat pads, and purple abdominal striae. Medications include metoprolol, lisinopril, amlodipine, metformin, insulin, and oral contraceptive pills.
    a. What clinical syndrome should you be concerned about?
    b. What laboratory tests should you order and why?
    c. What imaging studies are indicated?
    d. What is the treatment of choice for this syndrome?

2. A 60 year-old male is referred to an endocrinologist after a 6mm pituitary adenoma was discovered on an MRI done after he hit his head in a car accident. He reports no headaches or vision problems. In addition, he reports no problems with erectile dysfunction, libido or weight gain. He has not had any changes in his ring or hat size. On exam, he does not have supraclavicular fat pads, thin skin, or abdominal striae. There is also no frontal bossing and his teeth are straight.
    a. Are any laboratory tests needed? If so, what are they?
    b. How should his pituitary adenoma be followed clinically?

3. A 27 year-old female presents to her primary care physician with amenorrhea. She reports that menses commenced at age 12 and she had regular periods up until seven months ago when they became irregular. She now has not had a period in three months. The primary care doctor ordered a prolactin level that was found to be elevated at 120 ng/ml.
    a. What additional information should be elicited from the history?
    b. What are the next steps in the management of this patient?

Pituitary Anatomy

Located at the base of the skull in the sella turcica (part of the sphenoid bone).

- Critical structures nearby:
    - Optic chiasm- superiorly
    - Cavernous sinuses- laterally, contain CN III, IV, V, VI
    - Carotid arteries

Pituitary Imaging

- Basic principal of endocrinology- get the biochemical diagnosis before you start looking for the source
- 10% of otherwise healthy people will have nonfunctional asymptomatic tumors (incidentalomas)
- Modern imaging techniques allow us to visualize the sella with great detail (CT scan, MRI). MRI is the most sensitive imaging modality.

Sellar Enlargement

- Causes
    - Intrasellar tumors (pituitary adenomas most common)

- Hypophysitis (inflammation)
- Parasellar tumor (glioma, meningioma)
- Intrasellar aneurysm, hemorrhage
- Infiltrative disease (sarcoid)
- Empty Sella Syndrome
    - Primary empty sella - defect in the diaphragm of the sella, thought to allow cerebrospinal fluid (CSF) to enlarge sella/ compress the pituitary
        - Usually no pituitary dysfunction
        - Sometimes elevates prolactin
    - Secondary empty sella - association of empty sella with an identifiable disease of the pituitary gland
        - Example: pituitary adenoma that enlarges the sella but then is removed by surgery, radiation
        - May have pituitary dysfunction
    - Evaluation of a patient whose MRI shows an empty or partially empty sella, depends on clinical situation….(ex. checking for hormone deficiencies in secondary, if any clinical symptoms)

Pituitary Hormones, Adenomas & Definition of Terms

Pituitary adenomas account for approx. 90% of sellar tumors, virtually always benign

- Microadenoma is a tumor <1cm; macroadenoma is >1cm
- Adenomas can be functional or nonfunctional (producing hormone or not)
- Both small and large tumors, if hormone secreting, can present with signs and symptoms of hormone excess
- Mass effect- *large* tumors can compress surrounding structures
    - Visual loss (bitemporal hemianopsia)
    - Cranial nerve palsies
    - Headaches
    - Large tumors can also cause loss of function of other pituitary hormones (compress or invade the rest of the normal gland) Panhypopituitarism is the loss of all pituitary hormonal function
    - Define whether the loss of hormone is primary, which means that the actual end-organ gland is the etiology (ex- primary hypothyroidism -the

thyroid *gland* is not producing enough hormone), or <u>secondary</u>- something other than the gland that is the etiology of the dysfunction (ex. secondary hypothyroidism- pituitary dysfunction alters TSH secretion)

Growth Hormone (GH)

Function- Induces the hepatic production of insulin like growth factor 1 (IGF-1), promotes growth. Pulsatile release, fluctuate throughout life

- GH Overproduction
  - Gigantism in children, Acromegaly in adults
  - Rare, almost always from pituitary adenoma (usually macro)
  - More rare: Ectopic GH or GHRH secretion (lung, breast, pancreatic carcinoid tumors)
- Acromegaly clinical signs and symptoms:
  - Insidious (mean time from hypersecretion of hormone to actual diagnosis is TEN YEARS!)
  - Enlarged hands and feet
  - Enlargement of facial bones (increased mandible (prognathism) and brow (supraorbital ridges)
  - Widening of the space between teeth
  - Sweating
  - Enlargement of the soft tissues (sleep apnea, carpal tunnel)
  - Cardiomegaly, hypertension
  - Colon polyps/ carcinoma?, skin tags
  - Diabetes
  - Menstrual abnormalities
  - Untreated acromegalics- *high risk* for cardiovascular disease
- **Diagnosis**
  - GH is pulsatile with fluctuating levels throughout life, low levels in adults, so the Screening lab test-is IGF-1 (doesn't fluctuate as much)
  - Confirm biochemical diagnosis by Glucose tolerance test (glucose should suppress GH secretion in healthy people)
    - 100 grams of oral glucose, measure GH at 60 minutes and 120 minutes, levels>1ng/mL- confirm hypersecretion)

- Once have biochemical diagnosis, then do imaging

- **Treatment-**
  - Surgery (Transsphenoidal approach if possible)
  - Medical treatment - somatostatin (octreotide, dopamine agonists, GH receptor antagonists)
  - Radiation (Used as adjunct treatment to surgery if tumor incompletely removed/ recurs)

ACTH (Adrenocorticotropic hormone)

Function- stimulates the adrenal cortex to release glucocorticoids (cortisol). Cortisol maintains carbohydrate, protein, and fat metabolism, physiologic integrity of the immune and circulatory systems. Released in pulsatile fashion and in circadian rhythm (highest levels early before awakening, lowest in the evening)

- ACTH overproduction will lead to Glucocorticoid excess, which leads to a constellation of symptoms and physical features known as CUSHING'S SYNDROME
- Clinical Signs and Symptoms:

  Buffalo hump, central obesity, round rosy face, supraclavicular fat pads, carbohydrate intolerance, hypertension, thin skin, prominent purple striae, easy bruising, acne, hirsutism, menstrual disturbances, hypokalemia, edema, muscle weakness, osteoporosis/osteopenia, emotional lability

- Diagnosis

  Confirmed with at least 2 positive screening tests showing cortisol excess:

  - 24 hour urinary free cortisol well above the physiologic range (>150mcg/day)
  - Elevated Late night Salivary Cortisol
  - Overnight low-dose dexamethasone suppression test

    1 mg dexamethasone at 11pm, draw cortisol at 8am, Patients with Cushings do not suppress their 8 am cortisol below 1.8ng/dL.

    10-20% false positive results (obesity depression alcoholism). Confirm abnormal tests

- Next- measure ACTH
  - Excess cortisol can be from pituitary, adrenals, or ectopic tissue!

- If high ACTH, this is ACTH dependent Cushings (if cortisol is elevated, expect negative feedback on ACTH, and ACTH will be low; if there is an ACTH secreting tumor, cortisol will be high AND the ACTH will be high)
    - If ACTH < 5 pg/ml => Adrenal Cushing's => CT scan of adrenals
    - If ACTH > 20 pg/ml => Pituitary or Ectopic Cushing's
    - If ACTH between 5 and 20 pg/ml => Not definitive
- It may be very difficult to distinguish the different causes of ACTH dependent Cushings
    - Cushing's Disease means source is pituitary tumor (80% endogenous causes)
    - Ectopic ACTH- nonpituitary tumor making ACTH
        - 5-10% cases non-iatrogenic Cushings, ACTH usually very high
        - 90%+ from chest tumors (bronchial carcinoid, small cell lung cancer) medullary thyroid cancer, islet cell tumors, pheochromocytoma
        - Can present with signs/ symptoms of carcinoma, do imaging to search for site (Chest CT)
        - Treatment- tumor resection

- How to Localize ACTH source?
    - High dose dexamethasone suppression test- expect pituitary source to suppress more- controversy over lab cut-off values false positives)
    - Inferior petrosal sinus sampling: pituitary drains into petrosal sinuses- sample blood directly, calculate ratio ACTH from pituitary, compare to periphery.

      Ratio > 3 consistent with pituitary cause (Cushings disease)

- Cushing's Disease (pituitary tumor)
    - Usually a microadenoma
    - Once you get biochemical diagnosis, do imaging (MRI)
    - Treatment

      Surgery (Transsphenoidal approach if possible)

Radiation (adjunct to surgery, tumor incompletely removed or recurs)

B/L Adrenalectomy (always irradiate the pituitary before this*)

Medical treatment-drugs that block adrenal glucocorticoid synthesis (ketoconazole, metyrapone), somatostatin analogue

TSH (Thyroid Stimulating Hormone)

Function-activates synthesis and release of thyroid hormone

- TSH overproduction- Secondary Hyperthyroidism
  - TSH secreting pituitary adenoma is EXTREMELY rare
  - Diagnosis
    - Signs and symptoms of hyperthyroidism
    - Labs: TSH is normal or slightly high and thyroid hormone levels are high (T4, T3), which is inappropriate
  - Treatment surgical resection of pituitary tumor, +/- radiation (if remove or destroy the thyroid gland, you can cause pituitary tumor growth)
  - These tumors can co-secrete other hormones (FSH, LH)

FSH/LH (Gonadotrophins)

Function- bind to receptors in ovaries and testes- puberty, sex steroid production, reproductive function (ovulation)

- **Gonadotroph secreting tumor**
  - RARE, Usually macroadenomas
  - Clinical signs and symptoms-hypogonadism (loss of pulsatile secretion and complex regulation) impotence, infertility, loss of menses- rarely can cause increased sex steroids
  - Treatment is surgical resection +/- radiation

Prolactin

Different- tonically inhibited by dopamine, low secretion in 'normal' pituitary

Stimulus for release is pregnancy, TRH

Function is to stimulate milk production

Prolactinoma- most common *secretory* type of pituitary adenoma

- **Prolactin Hypersecretion**
  - Clinical Signs and symptoms: loss of menses (amenorrhea), galactorrhea (lactation), infertility, sexual dysfunction (loss of libido, impotence)
    - Prolactin inhibits pulsatility in gonadotroph secretion, leading to estrogen or testosterone deficiency
  - Differential diagnosis of high prolactin
    - Pregnancy
    - Renal failure
    - Primary hypothyroidism (TRH can stimulate release)
    - Microprolactinoma or macroprolactinoma
    - Chest wall stimulation, nipple stimulation
    - Medications (phenothiazines, metoclopramide)
    - Macroadenoma (other large pituitary tumor)
    - Stalk effect (anything disrupting the hypothalamic stalk can interrupt tonic inhibition of dopamine)
  - Diagnosis
    - Check prolactin, pregnancy test, TSH, review history for pituitary, renal or hepatic dysfunction, medication review
  - Treatment
    - Indications: hypogonadism, bothersome symptoms (galactorrhea), macroprolactinoma, infertility
    - Usually first line treatment is medical. Dopamine agonists (cabergoline, bromocriptine)cause tumor shrinkage, can restore menses and fertility
    - Surgery can be considered – but only with a highly experienced neurosurgeon and adenoma is well circumscribed (rarely done)
    - If fail to respond to medication or **severe** mass effect, then surgery is done

Non-Secreting (Null cell) Adenoma

Biologically inactive glycoprotein hormones may be secreted

Usually macroadenomas, can present with mass effect and compressive symptoms

Treatment is surgical +/- radiation

Posterior Pituitary

ADH- regulates water balance by binding to receptors in the renal tubules, facilitates water reabsorption

- 2022: An international group has recommended that the name "diabetes insipidus" -which in some cases has led to harm- be changed (eliminates confusion with "diabetes mellitus" and reflects pathophysiology)

  AVP-deficiency (central DI)

  AVP-resistance (nephrogenic DI)

- AVP Deficiency (Central Diabetes Insipidus)
  - Pituitary or hypothalamic loss of AVP
  - Have loss of water (Polyuria and Polydipsia, large volumes of **dilute** urine, craves cold water)
  - Causes: tumor, infiltrative disease, post-surgery, infection, familial, autoimmune, idiopathic
  - Treat by giving back AVP -causes the urine to concentrate again, dosed symptomatically by patients
  - If patients are able to drink and have access to water, they will be ok, but uncomfortable (constant urination and thirst)
- SIADH (Syndrome of inappropriate secretion of ADH)
  - Water retention, can lower serum sodium
  - Signs and symptoms- nausea, lethargy, seizures, coma
  - **Many causes (nausea, infection, medications, cancer)**; a hypothalamic tumor making ADH is extremely rare
  - Treatment- water restriction, or if severe, hypertonic saline

Hypopituitarism

- Clinical syndrome from deficiency of one or multiple pituitary hormones
- GH, FSH and LH are often lost first, TSH, ACTH and ADH are usually last
- Most common cause- pituitary macroadenoma

  Other Causes: Hypothalamic lesions (disruption of signals to pituitary) or pituitary destruction (neoplasm, infection, infiltration (granulomatous disease), hemorrhage, post-surgical, congenital or acquired lesions
- Will cause secondary endocrine gland failure (signals gone!)

- Pituitary Apoplexy- from hemorrhagic infarction of a pituitary adenoma
  - Acute presentation-Headache, vision loss, ophthalmoplegia, mental status change – may need surgical decompression
  - Workup-
    - Imaging (MRI pituitary)
    - Endocrine lab testing
    - Treatment- depends on etiology- may need surgery

- Hypopituitarism -Hormone replacement (see HPA lecture)
  - ACTH deficient- secondary Adrenal insufficiency, a medical emergency- replace glucocorticoid immediately
  - TSH deficient- thyroid hormone
  - LH/FSH deficient- estrogen or testosterone
  - GH deficient- GH
  - ADH deficient- give ADH (DDAVP)

- Prolactin Deficiency- failure to lactate post-partum
  - Sheehan's syndrome- postpartum hemorrhage into the sella Usually first sign of Sheehan's syndrome is inability to lactate- Must test all hormonal axes and provide replacement therapy (steroids most important -adrenal insufficiency) as needed

- Lymphocytic Hypophysitis
  - Lymphocytes infiltrate the pituitary / pituitary stalk-autoimmune
  - May not see mass on MRI (can see stalk thickening)
  - Usually seen in late pregnancy or postpartum-
  - Associated with hormonal dysfunction (loss of anterior pituitary hormonal function, diabetes insipidus)

Pituitary Incidentalomas

- Differential Diagnosis- see sellar enlargement above. Most commonly an adenoma
- Need to exclude hormonal hypersecretion:

Prolactin, IGF-1, TSH and Free T4, Screen for Cushings, LH, FSH

Treat accordingly

- If a Macroadenoma found- you also need to rule out hypopituitarism (hormone deficiency) and do formal visual field testing (optic chiasm nearby). If compression- surgery
- If all testing normal (nonfunctional, no compression danger) follow with periodic imaging an assessment
  - Macroadenoma- repeat MRI in 6 months
  - Microadenoma- repeat in 1 year, then 2 years, then 5 years if stable (very small <4 mm may not need repeat imaging)

Answers to Study Questions

These answers are not meant to be all encompassing; rather they are examples of the types of thought processes needed for application of the material.

1. The patient has physical exam findings concerning for Cushing syndrome. Although not every patient with Cushing syndrome has obvious exam findings, this patient does. Screening tests for Cushing's is indicated. Either 24-hr UFC and/or midnight salivary cortisol would be acceptable tests. The 1-mg overnight dexamethasone suppression test would not be the best test, since the patient takes oral contraceptive pills, which increases cortisol binding globulin (CBG) and could give a false-positive. Once hypercortisolemia is confirmed, ACTH will differentiate between pituitary/ectopic vs. adrenal Cushing. ACTH will be elevated or inappropriately normal in pituitary/ectopic Cushing and suppressed in adrenal Cushing's. Typically ACTH is much higher in ectopic Cushing's. Either MRI of the pituitary or imaging to look for ectopic tumors (usually lung) is indicated in ACTH dependent cases, and imaging of the adrenals is indicated in ACTH independent Cushing's. Surgery is the usual treatment of choice for all forms of Cushing's syndrome.

2. The patient has a 6mm pituitary incidentaloma with no signs or symptoms of hypersecretion. Despite this, it is recommended to rule out hypersecretion with laboratory tests. Prolactin, IGF-1, TSH, Free T4, LH, FSH, would be indicated. Although not all physicians screen for Cushing's in pituitary incidentalomas unless there are signs/symptoms of cortisol excess, ordering one of the screening tests would also be reasonable. Follow-up imaging should be done periodically (1 year, then 2 and 5 years) to

ensure stability in the size of the adenoma. If the adenoma was 10mm or greater, follow-up imaging should start at 6 months and testing for hypopituitarism would also be indicated.

3. The patient has hyperprolactinemia, but the etiology has not yet been elucidated. It is important to review medications that the patient is taking as atypical antipsychotics and other medications can elevate prolactin. She should also be asked about sexual history and the possibility that she could be pregnant, as well as any galactorrhea, visual problems and headaches. Other laboratory tests that are important are renal function, liver function, and thyroid function. She should have a pituitary MRI to look for a prolactinoma or a macroadenoma causing elevation of prolactin due to stalk compression. Treatment for a prolactinoma with amenorrhea and/or bothersome galactorrhea is a dopamine agonist (cabergoline or bromocriptine).